

Course: B. Tech.

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVE705D Rock Mechanics

Max Marks: 60

Date: 10/07/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Two of the following.

- | | |
|---|---|
| A) Write short notes on: - | 6 |
| a. Rock mass | |
| b. Primary objectives of studying rock mechanics | |
| c. How is rock mechanics used in the construction of tunnels? | |
| B) State and explain the principles and scope of rock mechanics. | 6 |
| C) Discuss the major milestones in the development of rock mechanics and their impact on engineering practices. | 6 |

Q.2 Solve Any Two of the following.

- | | |
|---|---|
| A) Write a short note on: - | 6 |
| a. What is flexural strength? | |
| b. Tensile strength of a rock | |
| c. What role does shear strength play in slope stability? | |
| B) Explain the procedures and equipment used in core drilling for rock sampling. | 6 |
| C) Describe the Mohr-Coulomb failure criterion and its application in rock mechanics. | 6 |

Q. 3 Solve Any Two of the following.

- | | |
|--|---|
| A) Write a short note on: - | 6 |
| a. Factors considered in RSR system | |
| b. Why is RQD important in rock mechanics? | |
| c. Name three basic types of rocks. | |
| B) Explain the procedure for determining the Rock Quality Designation (RQD) of a rock core sample. | 6 |
| C) Describe the main parameters considered in rock mass classification systems. | 6 |

Q.4 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. What factors influence the stability of rock slopes?
 - b. Why is monitoring important for rock slope stability?
 - c. How does planar failure occur?
- B) Describe the process and benefits of using grouting to improve rock properties. 6
- C) Discuss the various mechanical methods used to enhance the strength and stability of rock masses. 6

Q. 5 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. Various types of deformations in tunnels.
 - b. Type of rock support used in tunnels.
 - c. Method used to measure rock stresses around a tunnel.
- B) Describe the methods used to assess the suitability of a rock mass for foundation construction. 6
- C) Discuss the various types of tunnel linings and their functions. 6

***** End *****